

8 Artifacts and Technical Factors

Understanding and recognizing artifacts and technical factors play a central role in every nerve conduction and electromyography (EMG) study (Box 8.1). The value of the information gained during an electrodiagnostic (EDX) study relies on two important and complementary processes: (1) **collecting the data correctly** and (2) **interpreting the data correctly**. If the collected data are not technically accurate, then correct interpretation of the data can never occur, either at the time of the study or later by the referring physician.

EDX studies rely upon **acquiring and amplifying very small bioelectric signals** in the microvolt (μV) and millivolt (mV) range. This process is technically demanding, because a large number of physiologic and nonphysiologic factors can significantly interfere with the accuracy of the data. Physiologic factors, such as limb temperature and age, and nonphysiologic factors, such as electrode impedance and electrical noise, are equally important. Failure to recognize these technical factors that influence the EDX study can result in type I errors: diagnosing an abnormality when none is present (i.e., convicting an innocent man) and type

II errors: failing to recognize an abnormality when one is present (i.e., letting a guilty man go free). Although both are important, type I errors are potentially more serious. For example, “abnormalities” on EDX testing due to unrecognized technical errors can result in a patient being misdiagnosed with a condition they do not have. Such faulty diagnoses can lead to further inappropriate testing and treatment. Recognizing technical factors and other potential sources of error in the EDX laboratory is essential to improving the efficiency and validity of the EDX study and reducing patient discomfort.

PHYSIOLOGIC FACTORS

Temperature

Temperature is the most important of all the physiologic factors. It affects nearly every parameter measured in a nerve conduction study, including conduction velocity, distal latency, and waveform morphology. Temperature also can affect motor unit action potential (MUAP) morphology during the needle EMG examination. *Physiologically, cooler temperatures result in delayed inactivation of sodium channels and subsequently prolong the time of depolarization* (see Chapter 2). For myelinated fibers, conduction velocity is primarily determined by the time delay of depolarization that occurs at the nodes of Ranvier. Hence, prolonged depolarization times result in slowed conduction velocities for the nerve being studied. Conduction velocity slows in a fairly linear manner within the normal physiologic range of limb temperature (approximately 21–34°C). *For motor and sensory conduction velocities, conduction velocity slows between 1.5 and 2.5 m/s for every 1° C drop in temperature, and distal latency prolongs by approximately 0.2 ms per degree.*

In addition, longer channel opening time results in a larger influx of sodium. Subsequently, each nerve fiber depolarization is larger and longer. For both compound muscle action potentials (CMAPs) and sensory nerve action potentials (SNAPs), cooling results in a higher amplitude and longer duration as a consequence of larger and longer individual muscle and sensory fiber action potentials, respectively (Fig. 8.1). This effect is more pronounced in sensory fibers, because the duration of individual sensory nerve fiber action potentials normally is shorter than that of individual muscle fibers. The normal process of phase cancellation is more prominent when individual fiber action

Box 8.1 Important Technical Factors Influencing Nerve Conduction Studies and Electromyography

Physiologic Factors

- Temperature
- Age
- Height
- Proximal versus distal nerve segments
- Anomalous innervations (see Chapter 7)

Nonphysiologic Factors

- Electrode impedance mismatch and 60-Hz interference
- Filters
- Electronic averaging
- Stimulus artifact
- Cathode position: reversing stimulator polarity
- Supramaximal stimulation
- Co-stimulation of adjacent nerves
- Electrode placement for motor studies
- Antidromic versus orthodromic recording
- Distance between recording electrodes and nerve
- Distance between active and reference recording electrodes
- Limb position and distance measurements
- Limb position and waveform morphology
- Sweep speed and sensitivity

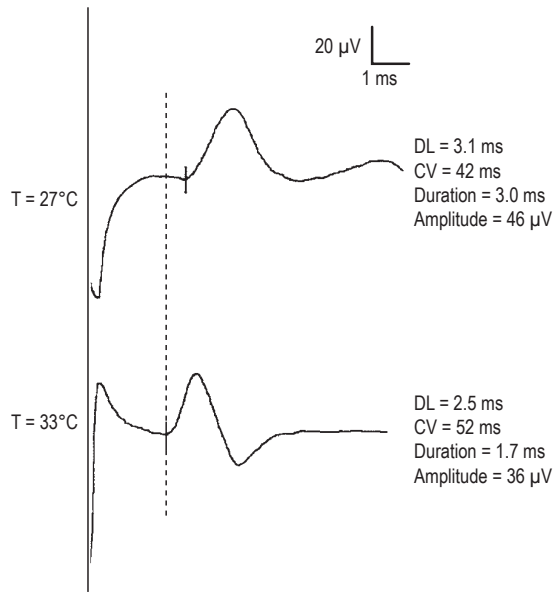


Fig. 8.1 Temperature effect on nerve conduction studies. Median antidromic sensory studies, stimulating wrist, recording second digit. Same patient at different limb temperatures. Note that with cooler limb temperature (*top*), distal latency (DL) and conduction velocity (CV) slow, while duration and amplitude increase.

potential durations are shorter (see [Chapter 3](#)). Thus, when cooling occurs, individual sensory nerve fiber action potentials prolong, resulting in less phase cancellation and a higher compound nerve action potential. Therefore cooling results in a very unusual pattern: slowed velocities but with increased amplitudes. In almost any neuropathy, slowing of conduction velocity is accompanied by *reduced* amplitudes. Accordingly, any sensory study that yields a high-amplitude, long-duration potential along with a slow conduction velocity should alert the electromyographer to a possible cooling effect.

Lowering the temperature has a similar though less marked effect on MUAPs measured during needle EMG. MUAP duration and amplitude increase with cooler temperatures; correspondingly, the number of phases also may increase.

There may be significant variation in limb temperature among individuals, even when ample time is allowed for temperature to equilibrate in a warm laboratory. Moreover, there is marked variation of temperature over the course of a given nerve, with a trend toward cooler temperatures as the nerve travels distally and superficially within the respective limb. Furthermore, in a warm limb, skin surface temperature typically is 1–2°C warmer compared with the near-nerve temperature. The opposite often is true in a cool limb, where the skin surface temperature typically is cooler compared with the near-nerve temperature.

It is easy to see how an electromyographer might mistakenly interpret a nerve conduction or needle EMG study as abnormal if a cool limb temperature is not appreciated and corrected. For example, a common mistake is to make a diagnosis of a polyneuropathy based on findings of slowed conduction velocities, prolonged distal latencies, and slightly large, polyphasic MUAPs that actually are due to

cold limb temperatures. Another common misdiagnosis is that of a distal entrapment neuropathy. For example, prolonged distal median motor and sensory latencies in a cool limb may create the false impression of a median neuropathy at the wrist (i.e., carpal tunnel syndrome). Lastly, in patients with axonal peripheral neuropathies, cooling may slow nerve conduction velocity into the range associated with demyelination, which then could profoundly alter the EDX impression and subsequent evaluation and treatment.

There are several ways to reduce the influence of cooling on the EDX study. First, the electromyographer must recognize the importance of temperature in every nerve conduction and EMG study ([Box 8.2](#)). Distal limb temperatures should be routinely recorded and monitored in all patients and ideally maintained between 32 and 34°C. A normal temperature at the beginning of a study does not ensure that the limb will not cool down as the study progresses; in fact, it often does cool down.

Limbs can be heated with heating lamps, warming packs, or hydrocollators. The ideal way to warm and maintain proper limb temperature is to use a heating lamp device that has a feedback control mechanism from a temperature sensor placed on the distal hand or foot. Unfortunately, these devices are now very difficult to obtain because most manufacturers have stopped producing them due to concerns over litigation; there were incidents where patients were burned when they grabbed the heating element and did not realize it was a heating device. It is important to be aware, however, that regardless of the method used to warm the limb, there may be a significant delay between the time that the skin reaches the desired temperature and when the underlying nerve and muscle warm up. When a limb is warmed, skin temperature usually reaches the desired temperature several minutes before the underlying nerve and muscle do so.

Box 8.2 Temperature and Nerve Conduction Studies and Electromyography

Effects of Cool Temperature

- Slowed nerve conduction velocity
- Prolonged distal latency
- Increased amplitude and duration of potentials on nerve conduction (SNAP > CMAP)
- Increased duration, amplitude, and phases of MUAPs

Maintenance of Temperature

- Measure distal limb temperature in all patients
- Maintain temperature at 32–34°C with heating lamp, warm packs, or hydrocollator
- Remember there may be a delay when heating between when the skin and the underlying nerve reach the desired temperature
- If limb is profoundly cool (>10°C cooler than desired), immerse the limb in warm water and then maintain temperature with a heating lamp
- If limb cannot be warmed, use conversion factors of 1.5–2.5 m/s/°C for conduction velocity and 0.2 ms/°C for distal latency

CMAP, Compound muscle action potential; MUAP, motor unit action potential; SNAP, sensory nerve action potential.

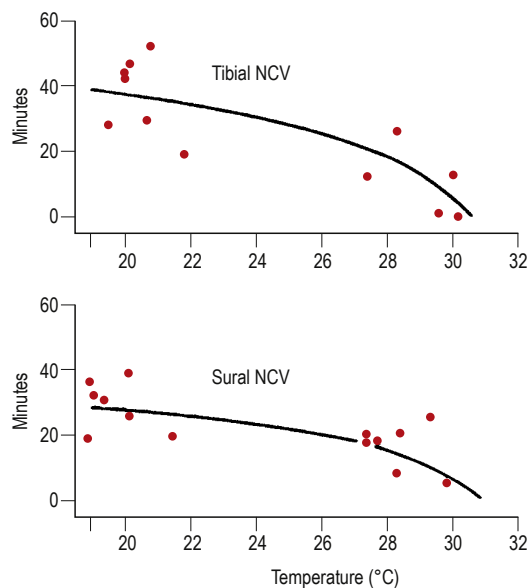


Fig. 8.2 Warming time and conduction velocity. The time required for conduction velocity of the tibial (upper) or sural (lower) nerve to reach 95% of its limit value is plotted against the skin temperature prior to warming. Note that even at a skin temperature of 28°C, 15–20 minutes may be required for nerve temperature to reach a steady state. NCV, Nerve conduction velocity. (Reprinted from Franssen H, Wieneke GH. Nerve conduction and temperature: necessary warming time. *Muscle Nerve*. 1994;17:336–344, with permission of Wiley.)

For limbs that are profoundly cool, it may require 20–40 minutes for the underlying nerve temperature to equilibrate (Fig. 8.2). If this fact is not recognized, the conduction velocity may increase through the EDX study as more time passes and the nerve warms up, despite a constant skin temperature. Nerves studied earlier in the examination will conduct more slowly than those studied later in the examination, after the underlying nerve has warmed up. This can result in confusing and difficult-to-interpret results. More often, however, the nerves cool down during the course of the study, and nerves studied later in the examination will conduct more slowly than those studied earlier in the examination.

If a limb is profoundly cool (e.g., more than 10°C cooler than desired), surface heating usually is inadequate or requires too much time to heat the limb adequately. In such situations, it is useful to immerse the limb in warm water and allow it to heat up over several minutes. Once the target temperature is reached, the proper temperature should be maintained; otherwise, the limb usually will cool down again during the course of the study.

The electromyographer must always keep in mind that a mildly to moderately slowed conduction velocity or a slightly to moderately prolonged latency could be the result of an initially cool limb or inadequate heating. If limb warming is not possible or is difficult to achieve (e.g., portable studies in the intensive care unit), then a **correction factor** should be used in interpreting the study, most commonly 1.5–2.5 m/s/°C for conduction velocity and 0.2 ms/°C for distal latency.

Several modern EMG machines can monitor limb temperature and can be set to automatically report corrected values for conduction velocity and latency based on temperature. However, one should keep in mind that these correction factors are derived primarily from individuals with normal nerves, and as such they may not hold true for all diseased nerves. *Hence, it always is preferable to warm or rewarm a limb than to use a correction factor.* On needle EMG, there is no correction factor for MUAP duration, amplitude, or phases.

Age

Age most prominently affects nerve conduction velocity and waveform morphology at the **extremes** of age. One of the most important determinants of nerve conduction velocity is the presence and amount of **myelin**. The process of myelination is age dependent and begins in utero, with nerve conduction velocities in full-term infants approximately **half** those of adult normal values. *Accordingly, nerve conduction velocities of 25–30 m/s are considered normal at birth but would be in the demyelinating range for an adult.* Conduction velocity rapidly increases after birth, reaching approximately 75% of adult normal values by age 1 year and the adult range by age 3–5 years, when myelination is complete.

Conduction velocities decrease slightly with age in adults, most likely as a consequence of the normal loss of motor and sensory neurons that occurs with aging. This is more prominent for individuals older than 60 years, in whom conduction velocity decreases approximately 0.5–4.0 m/s per decade. The effect is slightly more pronounced for **sensory** fibers than for motor fibers.

For adults, one can use normal values based on age for nerve conduction velocities, which take into account these small age-related changes in conduction velocity. However, it is difficult to remember values from tables based on both the age range and the nerve being studied. More commonly, tables of normal control values provide a range of nerve conduction velocities, usually for subjects between ages **10** and **60**, which take into account the normal variability within that age range. Generally, the lower limits of normal are provided. Additional correction factors of **0.5–4.0** m/s per decade can then be used for older patients. For example, a normal median motor conduction velocity in the adult population is **50** m/s or higher. However, in a **90**-year-old patient, a median motor conduction velocity of **46** m/s would be considered normal, within the expected range for advanced age.

Age also has an effect on CMAP and SNAP amplitudes. Again, most tables of normal control values provide a range of amplitudes that take into account normal variability for subjects between the ages of 10 and 60. Assessment of distal lower extremity sensory responses in an elderly individual, especially the commonly recorded sural sensory response, can be difficult. The **sural** sensory response in particular is often a small potential, and it may be hard to elicit in some older individuals. SNAP amplitudes are known to

decrease substantially with advanced age. Some have estimated that the SNAP amplitude drops by up to 50% by age 70. Thus, very low amplitude or absent lower extremity sensory responses in patients of advanced age must always be interpreted with caution and not necessarily considered abnormal without other confirmatory data.

Age also affects many parameters on the needle EMG study. The most prominent effect is on MUAP duration. It is well known that as an individual ages, MUAP duration increases. In the years from birth through childhood, the duration increases due to the physiologic increase of muscle fiber and motor unit size as an individual grows. In later life, the normal aging process results in a slow dropout of motor units. Some normal reinnervation occurs to compensate, which results in slight prolongation of motor unit duration as the individual ages. For these reasons, it is important to compare MUAP durations on needle EMG with normal values based on age (see Chapter 15).

Height

Along with temperature and age, height also affects nerve conduction velocity. Taller individuals commonly have slower conduction velocities than do shorter individuals. This effect of nerve length also is reflected in the well-recognized finding that normal conduction velocities are slower in the lower extremities, where the limbs are longer, than in the upper extremities. For example, on average, the normal sural sensory conduction velocity typically is 5 m/s slower than the median sensory conduction velocity, and the peroneal and tibial motor velocities typically are 6–9 m/s slower than the median and ulnar motor conduction velocities.

Two separate factors likely account for the effect of height or limb length on conduction velocity. First, nerves taper as they proceed distally. In general, the taller the individual, the longer the limb and the more tapered the distal nerve. Because conduction velocity is directly proportional to nerve diameter, the more distally tapered nerves in taller individuals conduct more slowly. By the same reasoning, nerves in the leg conduct more slowly than those in the arm because of longer limb length and more distal tapering. Second, and not as well appreciated, is that limbs are cooler distally than proximally and that the legs generally are cooler than the arms. Thus conduction velocity slowing due to cooling is usually more prominent in the legs than in the arms.

Tables of normal nerve conduction values usually take into account the range of normal height. However, modifications must be made for individuals of **extreme** height, just as they are needed for extremes of age. In practice, the adjustment usually is **no more than 2–4 m/s** below the lower limit of normal. For example, for an individual who is 6 feet 10 inches tall, a tibial conduction velocity of 38 m/s (the normal lower limit of normal is 41 m/s) should be considered within the normal range because of the effect of height.

The effect of height is especially relevant to the interpretation of late responses (F responses and H reflexes). The circuitry of these responses extends twice the length of the limb for the F response and twice the length of the

proximal lower limb for the H reflex. Normal values of absolute latency for these potentials must be based on limb length or height (see Chapter 4). Failure to do so will result in erroneously labeling of taller individuals as having “abnormal” late responses. In some situations, however, the effect of height is not relevant, as when latencies are compared between a symptomatic and a contralateral asymptomatic limb.

Proximal Versus Distal Nerve Segments

Nerve conduction velocities vary between the distal and proximal segments of a limb, just as they vary with height, because of changes in nerve diameter and temperature. Proximal nerve segments tend to conduct slightly faster than distal segments in a normal individual. For example, the normal conduction velocity of the median nerve between the axilla and elbow is slightly faster than between the elbow and wrist. This is true for the same reasons that conduction velocity is faster in the arms than in the legs: (1) distal nerve segments are tapered and therefore conduct more slowly than proximal segments and (2) distal limb segments are cooler than proximal segments and therefore conduct more slowly.

NONPHYSIOLOGIC FACTORS

Electrode Impedance and Noise

Electrical noise is present in every EDX laboratory. The most common cause of electrical noise is 60-Hz interference generated by other electrical devices (e.g., lights, fans, heaters, computers). Outside of the EDX laboratory, particularly in the intensive care unit, there may be many other sources of electrical noise such as ventilators, monitors, and other electrical devices. This noise can create a host of problems, especially when recording very small potentials such as SNAPs or fibrillation potentials (Fig. 8.3). However, the examiner usually can reduce electrical noise to an acceptable level by paying close attention to technical details.

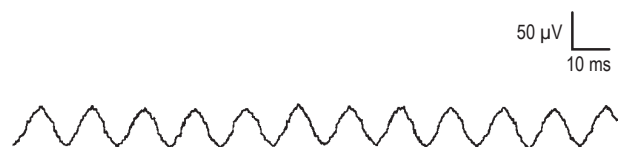


Fig. 8.3 Electrode impedance mismatch and electrical noise. Ambient 60-Hz electrical noise in the environment often can obscure small amplitude potentials (e.g., sensory nerve action potentials, fibrillation potentials). Electrical noise is recognized in the electromyography trace as a sinusoidal 60-Hz waveform. Note in the trace, the sweep speed of 10 ms allows one to appreciate the sinusoidal waveform. However, if the sweep speed is set at 1 or 2 ms and the sensitivity increased to 10 µV, as is done for sensory nerve conduction studies, the 60-Hz waveform can saturate the amplifier. The 60-Hz interference usually results from electrode impedance mismatch. If the impedances of the active and reference electrodes are similar, the same electrical noise is seen at the G1 and G2 inputs and subsequently is removed by differential amplification (common mode rejection). Electrode impedance mismatch can be reduced by proper skin cleaning and use of conducting electrode jelly.

All signals recorded during the nerve conduction study and needle EMG are the result of differential amplification (Fig. 8.4). With differential amplification, the difference between the signals at the active (G1) and reference (G2) electrodes is amplified and then displayed. Thus, if the *same* electrical noise is present at both the active and reference electrodes, it will be subtracted out, and only the signal of interest will be amplified (this is known as common mode rejection). However, if the electrical noise is larger at one of the electrodes than the other, the noise will not be subtracted out. When the electrical noise is amplified, it frequently is so large that it saturates the amplifier, resulting in a repeating waveform consisting of large vertical lines (Fig. 8.5, top). This waveform makes little sense until one changes the sensitivity to a much lower level (e.g., from 20 $\mu\text{V}/\text{div}$ to 10 mV/div), which discloses the source of the waveform: a 60-Hz sinusoidal waveform (Fig. 8.5, bottom). 60 Hz is the standard alternating current used in nearly all electrical devices in the United States; many other countries use 50 Hz.

The best way to achieve identical electrical noise at each electrode is to ensure that the impedance at each electrode is the same (i.e., to prevent electrode impedance mismatch).

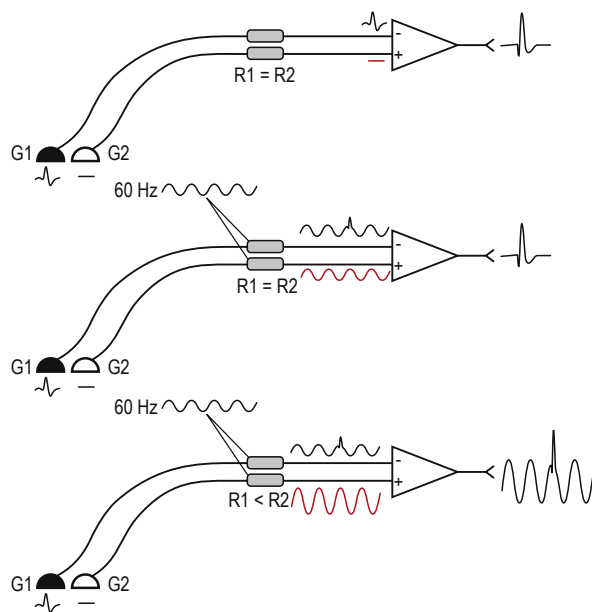


Fig. 8.4 Differential amplification and electrode impedance mismatch. All signals recorded in nerve conduction studies and electromyography result from differential amplification. *Top*, The signal present at the reference electrode (G2) is subtracted from the signal seen at the active electrode (G1) and amplified. Each recording electrode has its own impedance or resistance, modeled as $R1$ and $R2$, for the active and reference electrodes, respectively. *Middle*, If $R1$ and $R2$ are identical, any 60-Hz interference will induce a similar electrical noise at both inputs. The noise will then be subtracted out, and only the signal of interest will be amplified. *Bottom*, If electrode impedances are mismatched ($R1 < R2$), the amount of electrical noise will be different at the two inputs. Some of the electrical noise will then be amplified, often obliterating or obscuring the signal of interest.

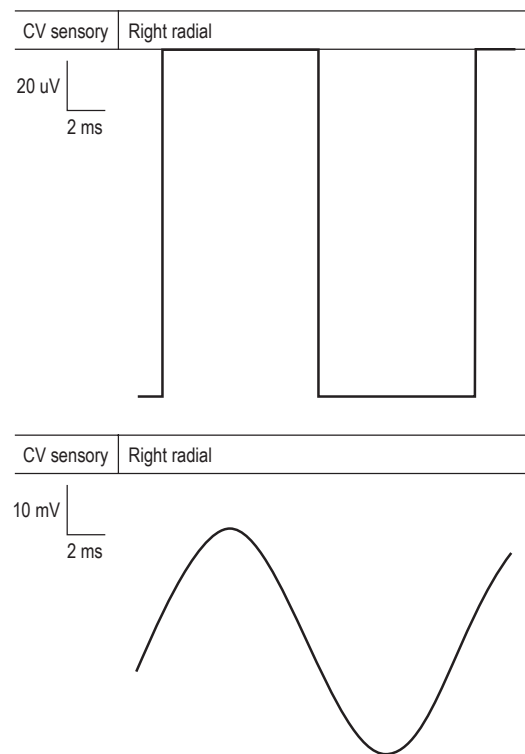


Fig. 8.5 Electrode impedance mismatch. Radial sensory study. Electrical noise is present but different at the two recording electrodes. This occurs due to electrode impedance mismatch. In this example, the resistance at one electrode is higher because of insufficient electrode gel. Because the electrical noise is not the same at the two electrodes, it will not be subtracted out when amplified. Sometimes, the resulting artifact is so large that it saturates the amplifier, resulting in a repeating waveform consisting of large vertical lines (*top*). This waveform is not recognizable until one changes the sensitivity to a much lower level (from 20 $\mu\text{V}/\text{div}$ [*top*] to 10 mV/div [*bottom*]), which discloses the source of the waveform: a 60-Hz sinusoidal alternating current waveform.

Impedance is an electrical term that combines the effects of resistance to flow for a direct current (DC) and capacitance and inductance for an alternating current. One will recall from Ohm's law that $E = IR$. The voltage (E), in this case the voltage from electrical noise, equals the current (I) induced from the electrical noise multiplied by the resistance (R), or impedance. If the resistance, or impedance, is different at the two electrodes, the same electrical noise will induce a different voltage at each electrode input. This difference will then be amplified and displayed, often obscuring the signal of interest.

The best way to eliminate 60-Hz interference is to ensure that each electrode appears identical to the amplifier (Box 8.3). This can be accomplished by several steps. First, ensure that the electrodes are intact, without any frayed or broken connections (Fig. 8.6). Next, the skin preparation should be thorough, using either alcohol or acetone to remove dirt and oil. Conducting electrode jelly is then applied to the electrode before it is attached to the skin.

Box 8.3 Methods to Reduce Electrode Impedance Mismatch and 60-Hz Interference

Active and reference recording electrodes should be the same type.
 Ensure all contacts are intact without any frayed or broken connections.
 Clean all dirt and oil from the skin using alcohol or acetone.
 Apply conducting electrode jelly between the skin and electrodes.
 Secure electrodes firmly to the skin with tape or Velcro straps.
 Place ground between stimulator and recording electrodes.
 Use coaxial recording cables.

The recording electrodes should be held firmly against the skin with tape or a Velcro band. Finally, the closer the electrodes are to each other, the more likely any associated electrical noise will appear identical to a differential amplifier.

Filters

Every potential recorded during nerve conduction studies and needle EMG passes through both a low- and a high-frequency filter before being displayed. The role of the filters is to faithfully reproduce the signal of interest while trying to exclude both low- and high-frequency electrical noise. Low-frequency (high-pass) filters exclude signals below a set frequency while allowing higher-frequency signals to pass through. High-frequency (low-pass) filters exclude signals above a certain frequency while allowing lower-frequency signals to pass through. Low-frequency noise (<10 Hz) results in wandering of the baseline (close to DC), whereas high-frequency noise (>10 kHz) commonly obscures high-frequency potentials (e.g., SNAPs or fibrillation potentials).

By allowing the signal to pass through a certain “passband,” some unwanted electrical noise can be excluded (Fig. 8.7). The passband varies for different EDX studies. For motor conduction studies, the low- and high-frequency filters typically are set at 10 and 10 kHz, respectively. For sensory conduction studies, the low- and high-frequency filters typically are set at 20 and 2 kHz, respectively. Note that the high-frequency filter is set lower for sensory than for motor nerve conduction studies. This is done to reduce high-frequency noise, which more easily interferes with the recording of SNAPs, which contain higher-frequency components compared with motor potentials (Fig. 8.8).

The use of filters always involves some tradeoff. No filter, whether analog or digital, results in a sharp cutoff with complete exclusion of all signals above the high-frequency settings or below the low-frequency settings. It is essential to recognize that filtering also results in some loss or alteration of the signal of interest. For instance, as the low-frequency filter is reduced, more low-frequency signals pass through. This results in the duration of the recorded potential increasing slightly because the duration is primarily a lower frequency response. Likewise, as the high-frequency filter is lowered, more high-frequency signals are excluded.



Fig. 8.6 Frayed electrode cable. It is often assumed that the electrodes and their connections are in working order. However, with prolonged use, wires and connections may fray and break, especially at points where cables end. This can result in a host of technical artifacts. To add to the confusion, frayed wires can lead to intermittent artifacts depending on the position of the wire and whether the connection is temporarily intact, partially broken, or completely separated.

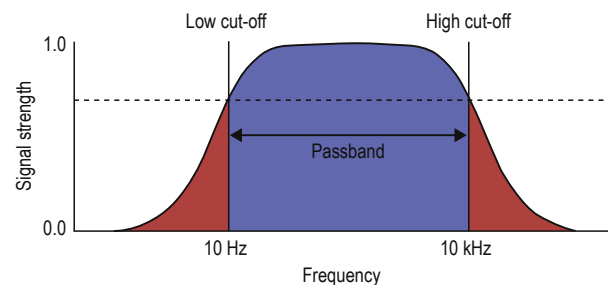


Fig. 8.7 Filtering and the passband. All waveforms recorded in electrodiagnostics are composed of a spectrum of frequencies. By allowing the signal to pass through a certain “passband,” some unwanted electrical noise can be attenuated. In this example for motor conduction studies, the low- and high-frequency filters typically are set at 10 and 10 kHz, respectively. Frequencies lower than 10 Hz and greater than 10 kHz are attenuated. This improves the quality of the recorded waveform by excluding low- and high-frequency noise. However, there is a tradeoff as some of the actual response is attenuated as well. For more information, see Chapter 42.

This results in the amplitude of the recorded potential usually decreasing because amplitude is primarily a higher frequency response (Fig. 8.9). Accordingly, all potentials should be obtained with standardized filter settings and should be compared only with normal values based on studies using the same filter settings.

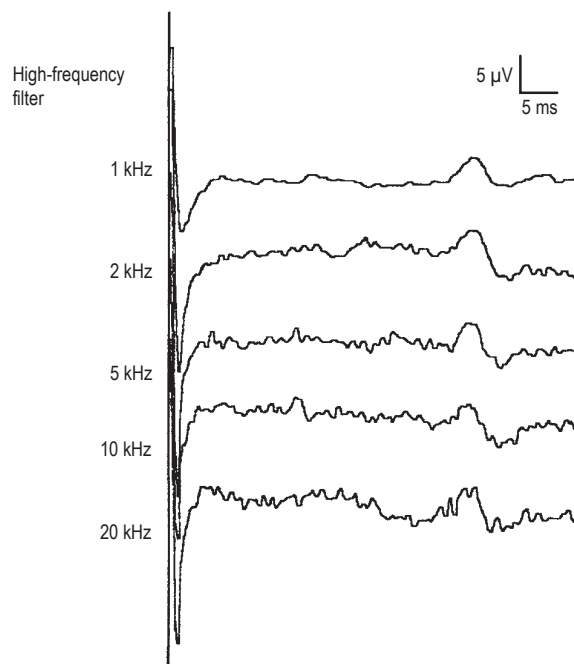


Fig. 8.8 High-frequency filter and sensory nerve action potentials. Ulnar sensory study, stimulating elbow, recording digit 5, with varying high-frequency filters. As the high-frequency filter is reduced from 20 kHz to 1 kHz, high-frequency noise is reduced and the sensory nerve action potential (SNAP) is better visualized. Note that the SNAP amplitude also decreases slightly.

Electronic Averaging

Electrical noise occasionally may contaminate a potential despite filtering and one's best efforts to eliminate electrode impedance mismatch. Most often this occurs when recording small potentials in the microvolt range, typically during sensory and mixed nerve studies. In this situation, the electrical noise can be reduced or eliminated by using electronic averaging. With electronic averaging, serial stimulations are digitized and then mathematically averaged. Because electrical noise is random, the positive and negative phases of electrical noise will cancel each other out as a greater number of stimulations are averaged, thereby leaving the potential of interest. Electronic averaging is especially helpful in clarifying the electrical baseline so that onset latency and amplitude can be correctly measured (Fig. 8.10).

Stimulus Artifact

During routine nerve conduction studies, the current from the stimulator depolarizes the underlying nerve, but it also spreads via volume conduction through the tissues within the limb and is seen at the recording electrodes. This stimulus artifact occurs in every nerve conduction study and serves a useful purpose by indicating when the shock occurred and from which point latencies should be measured. The stimulus artifact becomes problematic, however, if its trailing edge overlaps with the potential being recorded. This occurs

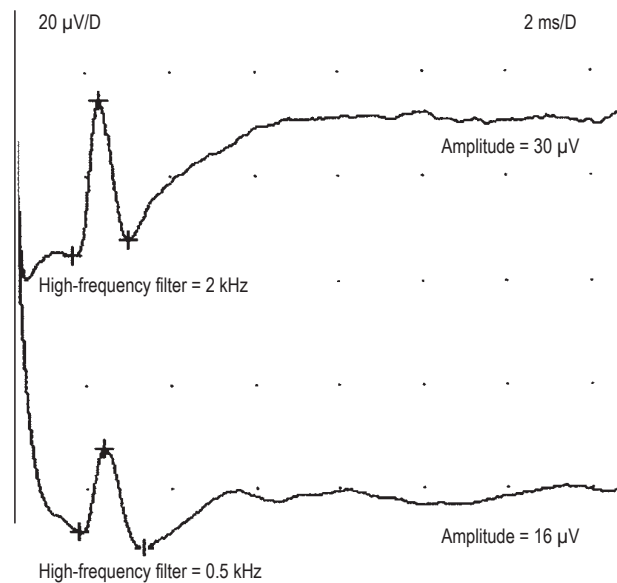


Fig. 8.9 High-frequency filter and sensory nerve action potentials. Median sensory study, stimulating the wrist, recording digit 2. *Top trace*, High-frequency filter is set at 2 kHz. *Bottom trace*, High-frequency filter is set at 0.5 kHz. Note that as the higher frequencies are filtered out (*bottom trace*), the amplitude of the sensory potential markedly decreases.

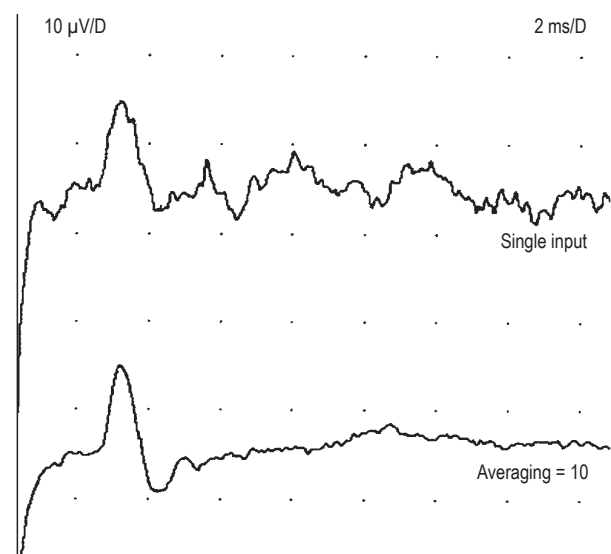


Fig. 8.10 Electronic averaging. Median sensory study, recording digit 2, stimulating the wrist. *Top trace*, Single stimulation. Note that the potential is present but that there is significant baseline noise. *Bottom trace*, Electronic averaging of ten stimulations. With the averaged trace, the noise is much improved, and the signal of the sensory response is more clearly seen, allowing the onset latency and amplitude to be more accurately measured.

most commonly when recording small potentials (i.e., sensory potentials) or when stimulating at very short distances. In those situations, the onset of the recorded potential may

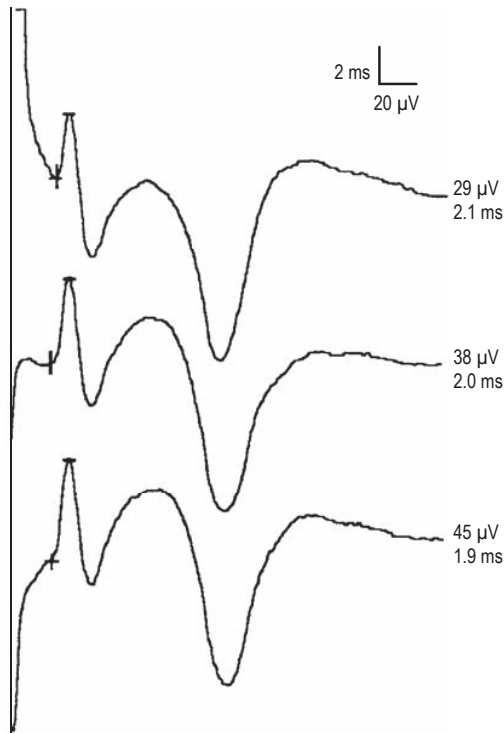


Fig. 8.11 Stimulus artifact and measurement error. Median antidromic sensory study, stimulating the wrist, recording second digit. Stimulus artifact can be influenced by rotating the anode while maintaining the cathode in place. Large negative stimulus artifacts (*top trace*) may result in artifactually low amplitudes and prolonged onset latencies. Conversely, large positive stimulus artifacts (*bottom trace*) may result in artifactually large amplitudes and short onset latencies.

Box 8.4 Methods to Reduce Stimulus Artifact

Place ground between stimulator and recording electrodes.
 Reduce electrode impedance mismatch between the recording electrodes.
 Use coaxial recording cables.
 Ensure stimulator position is optimized directly over the nerve.
 Lower the stimulus intensity.
 Rotate anode of the stimulator while maintaining the cathode.
 Increase the distance between the stimulator and recording electrodes.
 Ensure that the stimulator and recording electrode cables do not overlap.

be obscured, possibly leading to inaccurate measurements of both amplitude and latency (Fig. 8.11).

There are several ways to reduce the trailing edge of the stimulus artifact (Box 8.4). First, the optimal position for the ground electrode is between the stimulator and the recording electrodes to reduce stimulus artifact. Next, reducing electrode impedance mismatch between the recording electrodes will help reduce any electrical interference, including stimulus artifact. Coaxial recording cables are especially useful in this regard

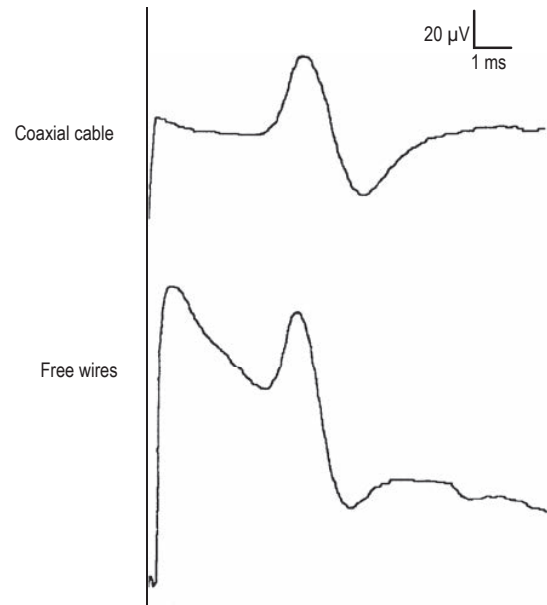


Fig. 8.12 Electrical noise and recording cables. Median antidromic sensory study, stimulating the wrist, recording digit 2 with ring electrodes, using different recording cables. Coaxial cable (*top trace*) and separate free wires (*bottom trace*). The closer the active and reference recording leads are to each other (coaxial cable closer than free wires), the less the chance that the stimulus artifact or any other electrical noise will be induced in the recorded trace.

(Fig. 8.12). Lowering the stimulus intensity also helps diminish the effects of the stimulus artifact. This is best accomplished by ensuring the stimulator is optimally placed on the nerve (see Chapter 3). The effect of the stimulus artifact also can be reduced by increasing the distance between the stimulator and the recording electrodes if possible.

Another very useful way to reduce the effect of the stimulus artifact is to slightly rotate the anode of the stimulator one way or another while maintaining the position of the cathode (known as walking the anode) (Fig. 8.13). Since the cathode position is unchanged, the depolarization still occurs at the same location. However, the geometric relationship between the axis of the cathode-anode and the recording electrodes can dramatically affect the direction and influence of the volume conducted stimulus artifact potential.

Finally, ensuring that the stimulator and recording electrode cables do not overlap and are kept as far apart as possible will help reduce the influence of the stimulus artifact. As will be discussed in Chapter 42 “Basics of Electricity and Electronics for Electrodiagnostic Studies,” overlapping cables can much more easily allow a stimulus artifact to occur in the recording electrode cables from the electrical property of induction. Keeping the stimulator cable away from the recording electrode cables is one of the simplest and most effective ways of reducing stimulus artifact.

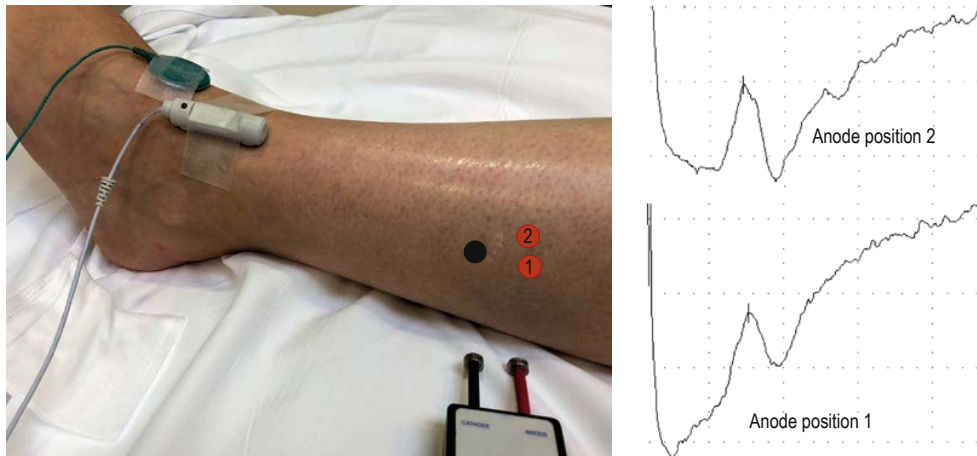


Fig. 8.13 Stimulus artifact and rotating the anode. Stimulus artifact often interferes with the potential of interest. One effective method to reduce stimulus artifact and obtain a clear baseline is to slightly rotate the anode of the stimulator one way or another while maintaining the position of the cathode (known as walking the anode). In this example, the superficial peroneal sensory nerve is stimulated in the lateral leg and recorded over the ankle. Note how the waveform changes as the anode is rotated between position 1 and position 2 while the cathode position is kept constant. In position 2, the waveform has a much improved baseline, which increases the accuracy of the onset latency and amplitude measurements. The stimulus artifact is a so called far-field potential. These potentials can be influenced by the geometry of the source in relation to the recording electrode.

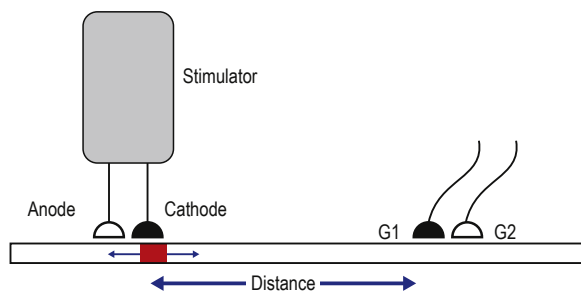


Fig. 8.14 Positions of the stimulator cathode and active recording electrode. With nerve stimulation, depolarization first occurs beneath the cathode and travels in both directions. The cathode should always face the active recording electrode (*G1*) (remember: black to black). To calculate a conduction velocity, distance is measured between the cathode and the active recording electrode.

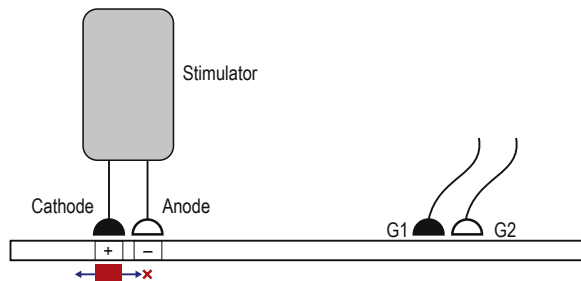


Fig. 8.15 Anodal block. If the cathode and anode are reversed, anodal block may occur. With stimulation, the nerve depolarizes beneath the cathode and travels in both directions, and the segment under the anode may hyperpolarize. This hyperpolarization may block the action potential that originates at the cathode and prevent it from proceeding past the anode.

Cathode Position: Reversing Stimulator Polarity

When a nerve is stimulated, depolarization first occurs beneath the cathode. Accordingly, distance measurements should always be made between the cathode of the stimulator, where depolarization first occurs, and the active recording electrode (Fig. 8.14). For nerve conduction studies, the proper position of the stimulating cathode is facing the active recording electrode (remember: black to black). If the cathode and anode of the stimulator are inadvertently reversed, two effects are possible. First, although depolarization occurs under the cathode, hyperpolarization theoretically occurs at the anode (Fig. 8.15). This hyperpolarization may create a block, preventing the depolarization that occurs under the cathode from proceeding past the anode to the recording electrode (i.e., anodal block). The resultant sensory or motor potential may then be reduced or absent. The issue of anodal block is more of a theoretic concern and is rarely seen in practice.

Second, and more common than a reduced or absent potential, is a predictable error in latency measurement when the cathode and anode are inadvertently reversed (Fig. 8.16). In such cases, the distal latency will prolong by approximately 0.3–0.4 ms. This represents the approximate time it takes for a normal nerve to traverse 2.5–3.0 cm, the typical distance between the cathode and anode of a stimulator. Failure to recognize this error results in an unusual set of findings. First, all the distal sensory nerve latencies will be prolonged by 0.3–0.4 ms, resulting in a slowing in sensory conduction velocities of about 10 m/s. Second, distal motor latencies will be similarly prolonged, but motor conduction velocities, which are calculated between a distal and a proximal site, will remain unchanged. The motor

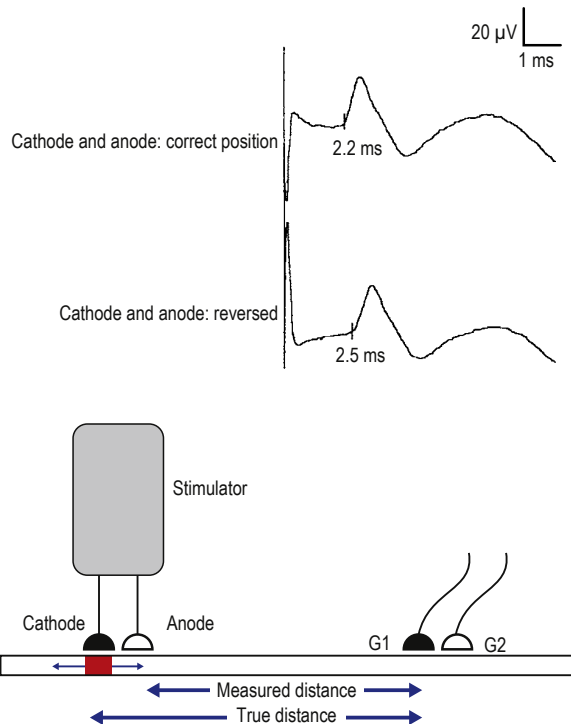


Fig. 8.16 Stimulator cathode and anode reversed. *Top*, Median sensory study, stimulating the wrist, recording digit 2. *Top trace* reflects the correct position with the cathode facing the active recording electrode. *Bottom trace* reflects the cathode and anode reversed with the cathode facing away from the active recording electrode. *Bottom*, If the cathode and anode are inadvertently reversed, artifactual slowing of latency and conduction velocity occurs. This error usually prolongs the latency by 0.3–0.4 ms and slows the sensory conduction velocity by approximately 10 m/s. The slowed conduction velocity is a result of the measured distance being shorter than the true distance traveled. The true distance is the measured distance plus the distance between the cathode and anode.

conduction velocities do not change because the distal latencies are subtracted out in the calculations. If the electromyographer does not recognize that the stimulating cathode and anode have been inadvertently reversed, these findings on the nerve conduction studies may easily be interpreted as consistent with a polyneuropathy or a distal entrapment neuropathy.

Inadvertently reversing the cathode and anode is not an uncommon mistake. It happens to junior as well as very senior electromyographers. Close attention to proper orientation of the cathode and anode is something that needs attention for every stimulation site.

Supramaximal Stimulation

Supramaximal stimulation is one of the most important concepts to understand when performing nerve conduction studies. All measurements made in nerve conduction studies are based on the assumption that all of the axons of the nerve have been depolarized. Different degrees of current intensity are required in different anatomic

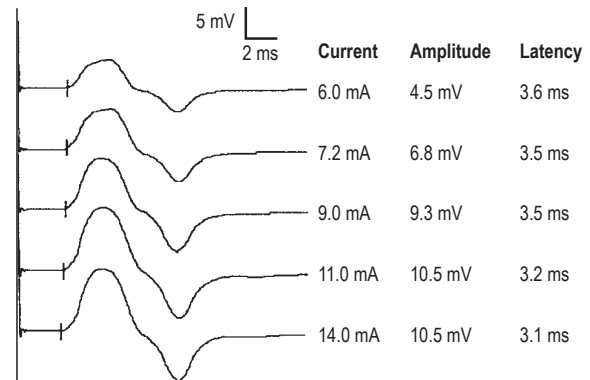


Fig. 8.17 Supramaximal stimulation. Median motor study, stimulating the wrist, recording the abductor pollicis brevis with increasing stimulator currents. To ensure that all axons are stimulated, supramaximal stimulation is required for all nerve conduction studies. Supramaximal stimulation is achieved by increasing the stimulator current until the recorded potential has reached maximal amplitude. To ensure supramaximal stimulation, the current should then be increased by an additional 25% to ensure that the amplitude does not increase further (*bottom trace*). Note that the latency also decreases as supramaximal stimulation is approached.

locations and in different individuals to depolarize all nerve fibers. For example, the current intensity needed to stimulate the median nerve at the wrist is much less than the current needed to stimulate the tibial nerve at the popliteal fossa.

To ensure that all axons have been depolarized, supramaximal stimulation must be performed. To achieve supramaximal stimulation, one must slowly increase current intensity until a point is reached where the amplitude of the recorded potential no longer increases. At that point, one increases the current an additional 25% to ensure that the potential will not change further. If it does not, then it can be assumed that supramaximal stimulation has been achieved (**Fig. 8.17**). Note that the latency decreases as supramaximal stimulation is approached.

If a nerve is not supramaximally stimulated at a distal site, a mistaken impression of axonal loss may result. If supramaximal stimulation is not achieved at a proximal stimulation site where it had been achieved with a distal stimulation site, the mistaken impression of a conduction block may occur (**Fig. 8.18**). In either case, depending on the nerve being stimulated, one might also mistakenly conclude that there is an anomalous innervation (see **Chapter 7**). Finally, a true conduction velocity cannot be obtained unless supramaximal stimulation is achieved at all stimulation sites. Conduction velocity measurements assume that the same fibers (including the fastest) are stimulated at distal and proximal sites. Without supramaximal stimulation, one may be measuring different fibers at different sites, resulting in invalid measurements of nerve conduction velocity. *One of the most common errors made in the EDX laboratory is to stop increasing the current once the amplitude of the potential falls within the "normal range."* In this case, the potential may be within the normal range, but

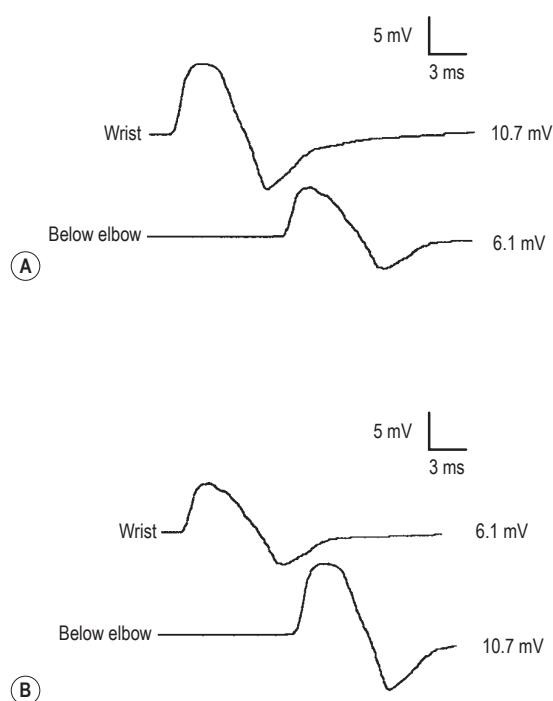


Fig. 8.18 Difference in compound muscle action potential (CMAP) amplitude between proximal and distal stimulations. Ulnar motor study, stimulating the wrist and below-elbow sites, recording hypothenar muscles. (A) CMAP amplitude is lower at the below-elbow site than at the wrist. This pattern may be seen with the following: (1) conduction block, (2) co-stimulation of the median and ulnar nerves at the wrist, (3) submaximal stimulation of the ulnar nerve at the below-elbow site, or (4) anomalous innervation (see [Chapter 7](#)). (B) CMAP amplitude is higher at the below-elbow site than at the wrist. This pattern may be seen with the following: (1) submaximal stimulation of the ulnar nerve at the wrist or (2) co-stimulation of the median and ulnar nerves at the below-elbow site. If this was a median motor study recording the abductor pollicis brevis, this pattern also could represent an anomalous innervation.

because it is not achieved with supramaximal stimulation, it may not be normal for the particular patient being studied.

Note that if one does not go through the meticulous process of incrementally increasing the stimulus current, one can never be certain that supramaximal stimulation has been achieved, no matter how high the stimulus current intensity! Thus, the electromyographer should never assume that the maximum stimulus machine output is the same as supramaximal stimulation without going through this process.

Co-Stimulation of Adjacent Nerves

Although it is imperative to ensure that supramaximal stimulation has been achieved at all stimulation sites, preventing co-stimulation of adjacent nerves is equally important. In individuals with normal nerves and normal stimulation thresholds, co-stimulation is not a common problem. In pathologic situations, however, nerves often require higher currents to achieve supramaximal stimulation. As the

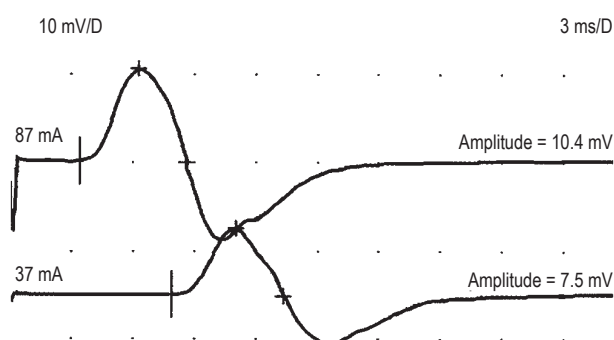


Fig. 8.19 Co-stimulation. Ulnar motor study, stimulating the wrist (top trace) and below-elbow (bottom trace) sites, recording the first dorsal interosseous muscle. Stimulus current intensities noted at the beginning of each trace. Note that the amplitude at the wrist site is significantly higher than at the below-elbow site. In this case, an error is caused by excessive current at the wrist that stimulates the ulnar nerve as well as the adjacent median nerve (i.e., co-stimulation). In general, for normal individuals, currents exceeding 50 mA with a duration of 0.2 ms often will result in co-stimulation of adjacent nerves. If not recognized and corrected, co-stimulation at a distal site may result in the mistaken impression of a partial conduction block at the proximal site or, in the case of the ulnar nerve, may mimic a Martin-Gruber anastomosis.

stimulus current is increased, the current may spread to excite nearby nerves. As nearby nerves are excited, spuriously large amplitude potentials may result, caused by the inadvertent co-recording of additional nerve or muscle potentials beyond the potential of interest. Co-stimulation occurs most commonly in motor studies of the upper extremity when the median and ulnar nerves are stimulated at the wrist, elbow, and axilla. In the lower extremity, co-stimulation of the peroneal and tibial nerves may occur at the knee.

Co-stimulation of adjacent nerves is unavoidable when stimulating very proximal nerves and nerve roots. In the upper extremity, stimulation at Erb's point or at the C8–T1 nerve roots always results in co-stimulation of both the ulnar and median nerves. In this situation, the effects of co-stimulation can only be eliminated by the use of collision studies (see [Chapter 33](#)).

Inadvertent co-stimulation of adjacent nerves can create a host of problems, even in routine nerve conduction studies ([Figs. 8.18 and 8.19](#)). First, a low-amplitude potential due to axonal loss may reach the normal range if an adjacent nerve is co-stimulated. Next, if co-stimulation occurs distally but not proximally, there may be the mistaken impression of a conduction block proximally (see [Fig. 8.20](#), top). In some nerves, such as the ulnar motor nerve, this pattern can also mimic an anomalous innervation. On the other hand, if co-stimulation occurs proximally but not distally, this pattern also can mimic an anomalous innervation in certain nerves, such as the peroneal motor nerve (see [Chapter 7](#)). Finally, if there is a true conduction block between the distal and proximal stimulation sites, co-stimulation at the proximal site will result in an inappropriately high amplitude proximally, which may obscure a true conduction block (see [Fig. 8.20](#), bottom).

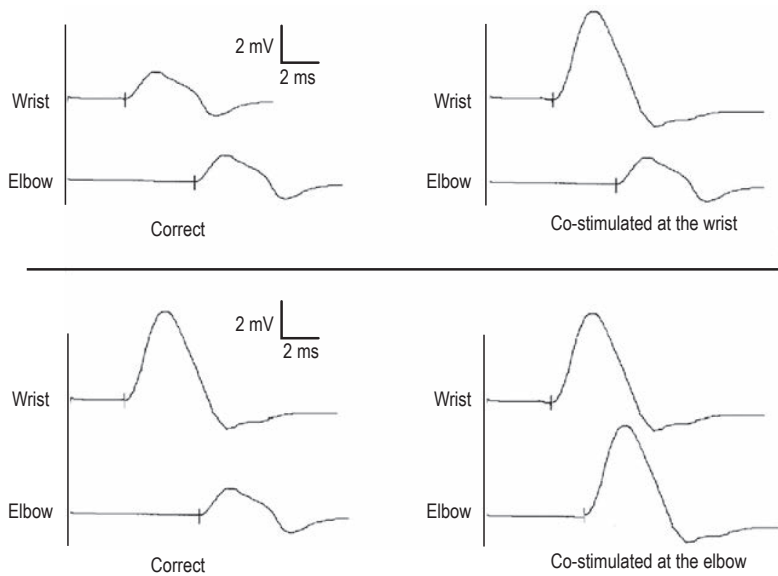


Fig. 8.20 Co-stimulation and interpretation of conduction block. During motor nerve conduction studies, conduction block is recognized by a marked drop of amplitude or area on proximal stimulation as compared to distal stimulation. However, if the nerve is co-stimulated at the distal or proximal sites, different problems occur. *Top, (left)* If the amplitude is truly reduced at the distal (wrist) and proximal (elbow) stimulation sites, the pattern is one of axonal loss. However, if the distal site is inadvertently co-stimulated (and not the proximal site) (*right*), then a spuriously higher amplitude response will be present distally, and one will erroneously make the electrical diagnosis of conduction block. *Bottom, (left)* If there is a true conduction block between distal and proximal stimulation sites, there will be a tendency to overstimulate the proximal site to get the amplitude “normal.” In this case, if the nerve is inadvertently co-stimulated at the proximal site (and not at the distal site) (*right*), the true conduction block pattern will be obscured, and the mistaken electrical diagnosis of a normal nerve conduction will be made.

Box 8.5 Methods to Avoid Co-Stimulation of Adjacent Nerves

Ensure stimulator position is optimized directly over the nerve.
Watch for abrupt change in waveform morphology.
Watch for a change in the resultant muscle twitch.
Avoid excessive stimulation currents.
If necessary, co-record muscles simultaneously from adjacent nerves.
Always consider possible distal co-stimulation when a conduction block is seen.

There are several ways to prevent co-stimulation of adjacent nerves (Box 8.5). First, co-stimulation often can be prevented by ensuring that the stimulator is placed directly over the nerve. By doing so, much less current is required to achieve supramaximal stimulation, and co-stimulation is easily prevented. The stimulator is placed over a site where the nerve is expected to run, based on anatomic landmarks. The stimulus intensity is slowly increased until the first small submaximal potential is recorded. At this point, the stimulus current is held constant, and the stimulator is moved parallel to the initial stimulation site, both slightly laterally and then slightly medially. The position that yields the highest-amplitude response is the position closest to the nerve. Once the optimal position is determined, the current is increased to supramaximal. It is often surprising how little current is required to obtain supramaximal stimulation using this technique, which also improves efficiency and patient tolerance of the procedure. Second, while watching the amplitude of the waveform increase as the stimulus intensity is increased, the shape of the waveform will often change abruptly when co-stimulation occurs. For instance, the normal dome shape of the median motor

response may abruptly develop a bifid morphology, signifying that the ulnar nerve now is being co-stimulated. Third, co-stimulation often can be prevented if attention is paid to the muscle twitch during stimulation. For example, stimulation of the median nerve at the wrist results in contraction of the thenar eminence and first two lumbricals. In contrast, ulnar nerve stimulation results in a more widespread flexion contraction of the hand as the ulnar nerve innervates most of the intrinsic hand muscles. Thus, as the current intensity increases and co-stimulation of median and ulnar innervated muscles begins, the observer will witness a change in the muscle twitch. At this point, the stimulus intensity should be decreased until the point that only the median innervated muscles contract. This also applies to the lower extremities, especially at the popliteal fossa where the tibial nerve is in close proximity to the peroneal nerve. Stimulation of the peroneal nerve results in ankle dorsiflexion and eversion, whereas stimulation of the tibial nerve results in ankle plantar flexion and inversion. Thus when stimulating the peroneal nerve at the knee, the normal twitch of ankle dorsiflexion will change to plantar flexion and inversion when the tibial nerve is co-stimulated. Finally, for most normal individuals, co-stimulation of the median and ulnar nerves at the wrist and elbow, and of the peroneal and tibial nerves at the lateral popliteal fossa, often occurs at stimulus intensities >50 mA (0.2 ms pulse duration). Thus once stimulus intensities are increased beyond this point, the electromyographer needs to appreciate the increased possibility of co-stimulation.

If there is still a question of co-stimulation after taking into account the previous suggestions, one should simultaneously record muscles innervated by adjacent nerves, watching for a potential from the unintended muscle. If such a potential occurs, the stimulus intensity should be

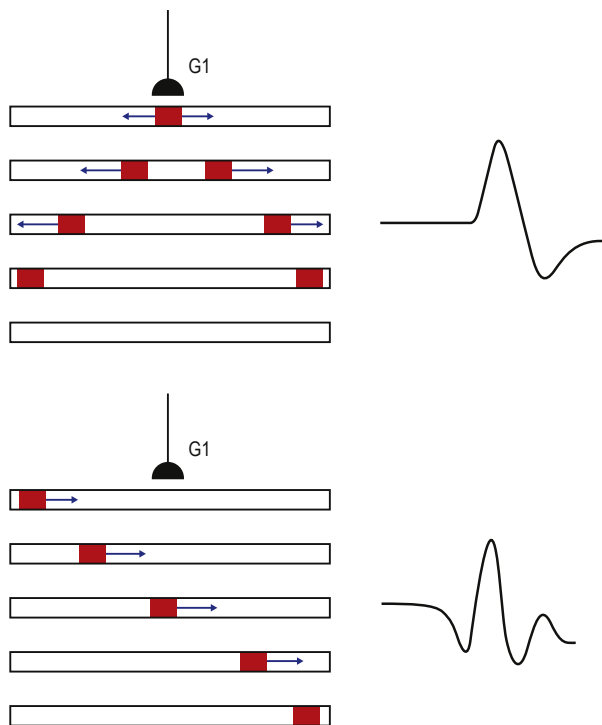


Fig. 8.21 Compound muscle action potential morphology and site of depolarization. Depolarization occurs first at the motor endplate (motor point), with the depolarization subsequently spreading away from that site. With the active recording electrode (*G1*) placed over the motor point, the corresponding waveform has an initial negative deflection without any initial positivity (*top*). If the active recording electrode is not placed over the motor point, depolarization begins at a distance from and then travels beneath and past the active electrode, resulting in an initial positive deflection (*bottom*).

lowered until the unintended potential is no longer seen. For instance, when stimulating the median nerve at the wrist, if there is a question of ulnar co-stimulation, the abductor pollicis brevis (median innervated) and abductor digiti minimi (ulnar innervated) should be simultaneously recorded. If median nerve stimulation is done correctly, no potential should be recorded from the abductor digiti minimi.

Electrode Placement for Motor Studies

The preferred montage for recording motor conduction studies is the belly-tendon method. The active electrode (*G1*) is placed over the motor point, typically located in the center of the muscle belly, while the reference electrode (*G2*) is placed over the muscle's distal tendon. When this montage is used, the muscle tendon site presumably represents an electrically inert point, and only the signal at *G1* is amplified.

Muscle depolarization occurs first at the motor endplate zone (motor point). If the active recording electrode is not placed over the motor point, the volume-conducted depolarization potential first occurs at a distance from the recording electrode and is seen as an initial positive deflection. When the depolarization subsequently travels under the electrode, the potential then becomes negative (Fig. 8.21). Two problems may occur with such an incorrect placement. First, the CMAP may not be maximized, giving the mistaken impression of a reduced amplitude (Fig. 8.22). Second, if an initial positive deflection occurs, the latency is difficult to measure (Fig. 8.23). Whenever an initial positive deflection is seen on a motor conduction study, the active electrode has most likely been placed off

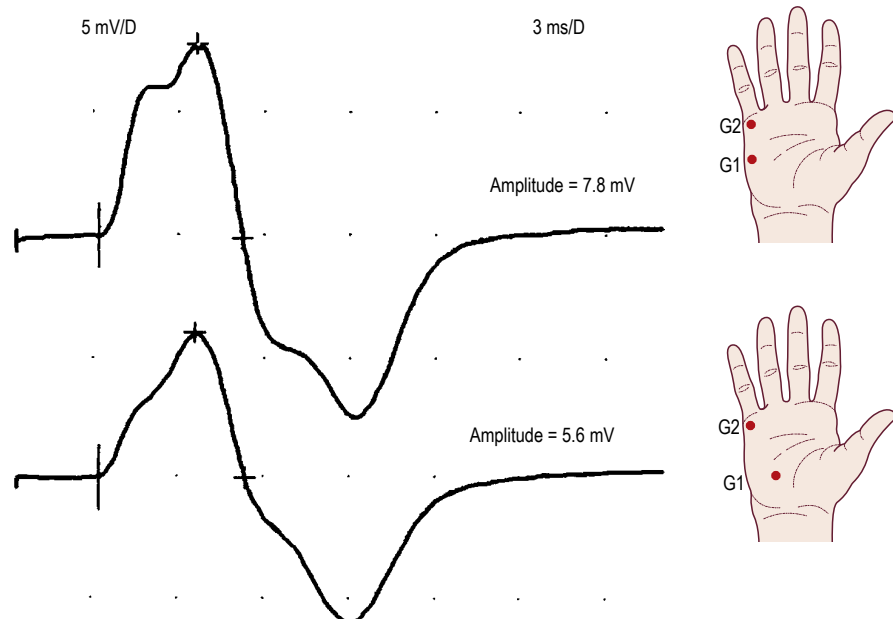


Fig. 8.22 Effect of active recording electrode position on amplitude in motor studies. Ulnar study recording the hypothenar muscles, stimulating the wrist. The optimal position to evoke the maximal amplitude is over the motor point (*top trace*). When the active recording electrode (*G1*) is off the motor point, a positive initial deflection often is noted, alerting the examiner to the incorrect placement. However, this may not occur, especially when nearby muscles also are depolarized (*bottom trace*). Repositioning the active recording electrode often may result in a higher amplitude. This is especially important when comparing potentials from side to side.

the motor point and should be moved until the positive deflection is no longer seen.

Not as well appreciated is the possibility of technical errors if the G2 electrode is misplaced. In the belly-tendon montage, it is generally assumed that the tendon is

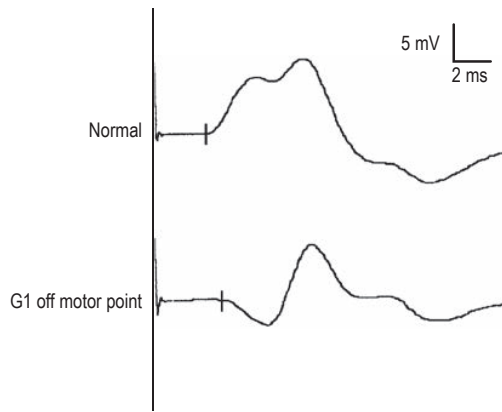


Fig. 8.23 Active recording electrode placement and motor studies. Ulnar motor study, recording the abductor digiti minimi, stimulating the wrist. The active recording electrode (G1) is properly placed over the motor point of the muscle, and the reference electrode (G2) is placed over the distal tendon (*top trace*). If G1 is placed off the motor point, the morphology of the compound muscle action potential changes, usually to show an initial positive deflection and a lower amplitude potential.

electrically inactive. Although this is true for most nerves, it is not so for all nerves, especially the ulnar and tibial nerves, where the reference electrode placed over the tendon is usually (and surprisingly) electrically active. Because there is no muscle over the tendon, this “tendon potential” is likely a volume-conducted far-field potential from nearby or proximal depolarizing muscles. In some cases, much of the CMAP amplitude is actually generated from the tendon potential (*Fig. 8.24*). These tendon potentials are predominantly positive. Thus, the depolarization from G1 (which is negative) minus the tendon potential from G2 (which is positive) usually creates a larger negative potential. The key to avoiding errors from different G2 locations is consistency. For instance, if the right ulnar nerve is studied with G2 placed at the base of the fifth digit but the left ulnar nerve is studied with G2 placed distally on the fifth digit, then different, asymmetric amplitudes may result, based solely on the difference in the position of G2 (*Fig. 8.25*).

Antidromic Versus Orthodromic Recording

For sensory conduction studies, either antidromic or orthodromic methods can be used. When a nerve is stimulated, conduction occurs equally well in both directions. Latencies and conduction velocities are identical using either method. However, each method has its advantages and disadvantages

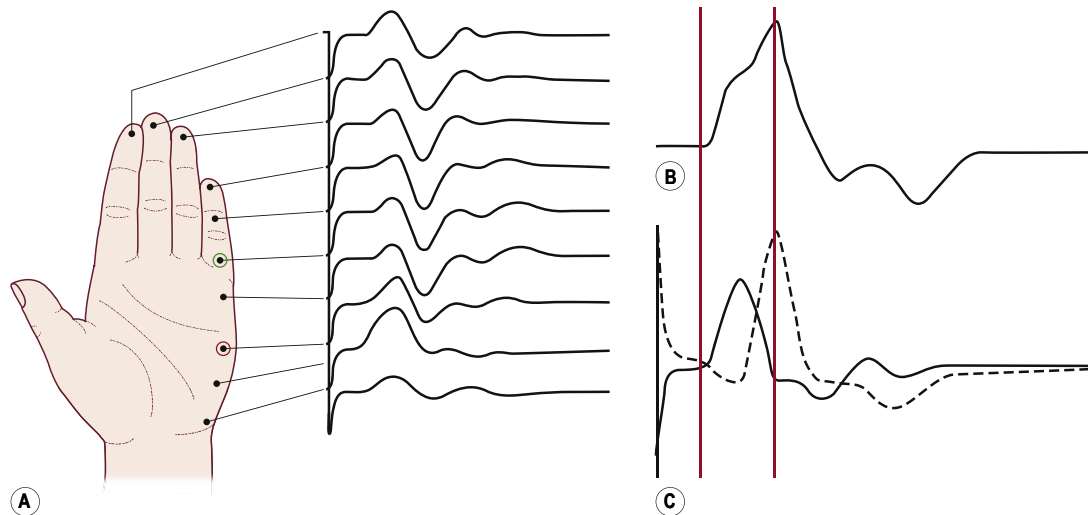


Fig. 8.24 Effect of the reference electrode on the compound muscle action potential (CMAP) configuration. Recording electrodes for motor studies are placed using the “belly-tendon” montage. The depolarization occurs under the muscle belly, where the active electrode (G1) is placed. However, the tendon may be electrically active, especially when studying the ulnar and tibial nerves. In this case, a tendon potential occurs as a result of volume conduction of proximal and other nearby potentials. In the case of the ulnar nerve, this gives the motor response its characteristic bifid morphology. To demonstrate this, look at all the waveforms in panel A. Each trace displays the ulnar CMAP when recording at a different active electrode site with the reference electrode on the contralateral hand. Note that position 3 (*red circle*) corresponds to the usual G1 site, and position 5 (*green circle*) to the usual G2 site for ulnar motor studies. Normally, G2 is subtracted from G1 and amplified to produce the final waveform. This is the same as adding the negative of G2 to G1. In panel C, the waveform with the solid line is from the standard G1 recording site; the waveform with the dashed line is the negative of the standard G2 or tendon site. When the two are added together, the final waveform is created in panel B. Note that most of the standard ulnar CMAP is actually created from the tendon potential. (Adapted from Kincaid JC, Brashear A, Markand ON. The influence of the reference electrode on CMAP configuration. *Muscle Nerve*. 1993;16(4):392–396.)

Fig. 8.25 Effect of reference electrode position on amplitude in ulnar motor studies. Standard ulnar motor studies with the G1 active electrode over the abductor digiti minimi while varying the position of the G2 reference electrode. The G2 reference electrode over the tendon is, in theory, electrically neutral. However, the tendon may be electrically active, especially when studying the ulnar and tibial nerves. Note in the three traces how the morphology and amplitude of the motor response change as the placement of the reference electrode is changed. This underscores the need for consistency in placing both the reference and active recording electrodes when performing motor studies.

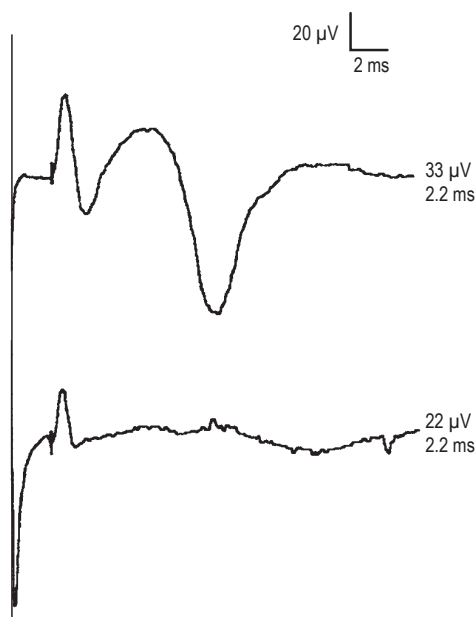
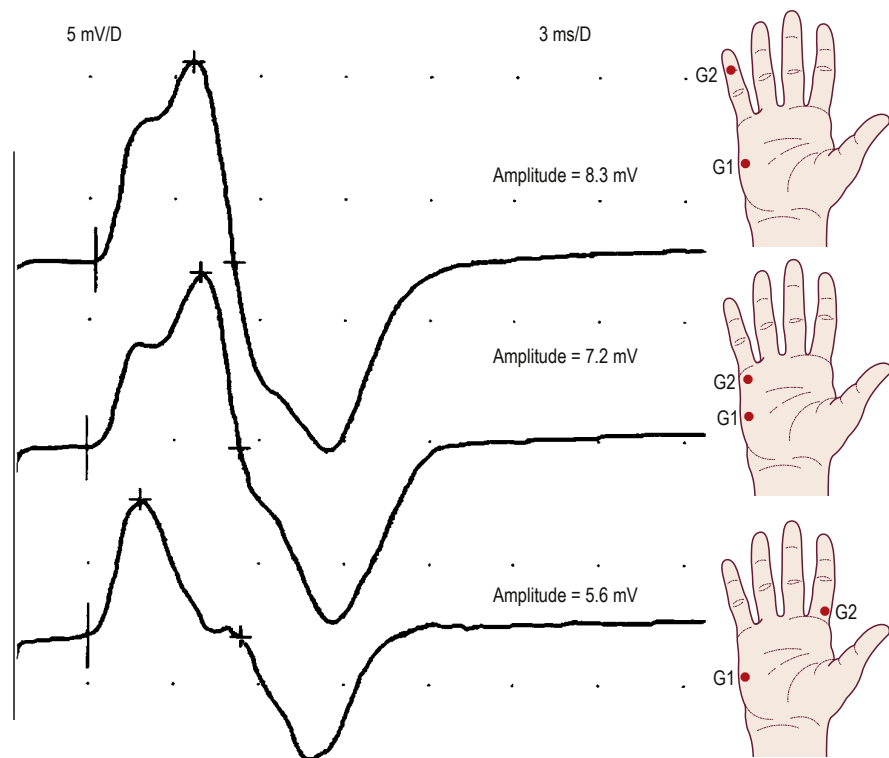


Fig. 8.26 Comparison of antidromic and orthodromic median sensory studies. *Top trace*, Antidromic study, stimulating the wrist, recording digit 2. *Bottom trace*, Orthodromic study, stimulating digit 2, recording the wrist, same distance. Latencies and conduction velocities are identical. The antidromic method has the advantage of a higher amplitude, but the sensory nerve action potential (SNAP) may be followed by a large volume-conducted motor potential. If the SNAP is absent in an antidromic study, one must be careful not to mistake the volume-conducted motor potential for a sensory potential.

(Fig. 8.26). First, amplitude is higher with antidromic than with orthodromic recordings. SNAP amplitude is directly proportional to the distance between the recording electrodes and the nerve. For most antidromic potentials, the active recording electrodes are closer to the nerve. For example, consider the antidromic median sensory study stimulating the wrist and recording the second digit. Using the antidromic method, recording ring electrodes are placed over the second digit. The ring electrodes are very close to the underlying digital nerves, which lie just beneath the skin. When the montage is reversed for orthodromic recording, the recording bar or disk electrodes are placed over the wrist. The thick transverse carpal ligament and other supporting connective tissue lie between the nerve and the recording electrodes. The recorded sensory response consequently is attenuated by the intervening tissue and results in a much lower amplitude. The major advantage of antidromic recording is the higher amplitude potentials obtained with this method. Not only is it easier to find the potential, but also larger amplitude potentials can be especially helpful in making side-to-side comparisons, following nerve injuries over time, or recording potentials from pathologic nerves, which can be quite small.

The antidromic method, however, does have its disadvantages. Although only sensory fibers are recorded, both motor and sensory fibers are stimulated. This often results in a volume-conducted motor potential following the SNAP (Fig. 8.26, and see Fig. 3.9). Because the SNAP usually occurs before the volume-conducted motor potential, it is not difficult to differentiate the two. However, if the two

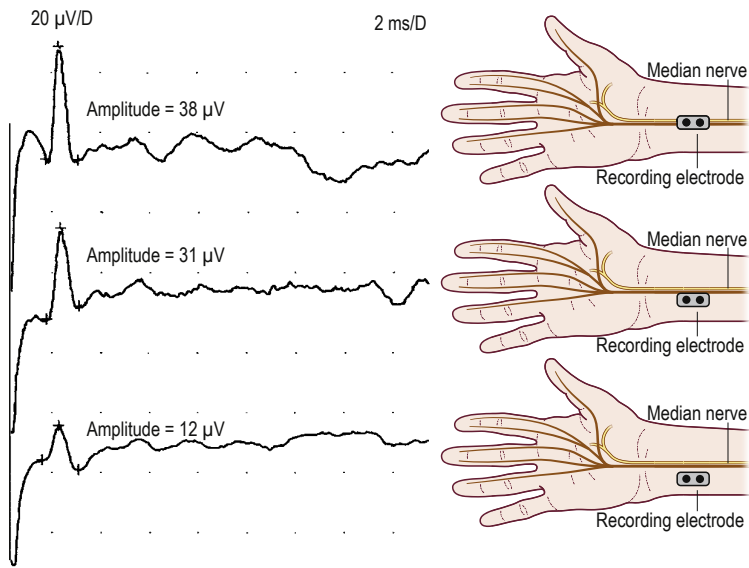


Fig. 8.27 Effect of distance between recording electrodes and nerve on amplitude. Median mixed nerve study, stimulating the palm, recording over the wrist. *Top trace*, Recording electrodes placed directly over the median nerve. *Middle trace*, Recording electrodes placed 0.5 cm laterally. *Bottom trace*, Recording electrodes placed 1.0 cm laterally. If the recording electrodes are moved off the nerve (middle and bottom traces), maintaining the same distance and stimulus current, the amplitude drops markedly. In nerve conduction studies, side-to-side comparisons between amplitudes are often made, looking for asymmetry. One can easily appreciate that if the recording electrodes are placed lateral or medial to the nerve on one side and directly over the nerve on the other side, one might be left with the mistaken impression of a significant asymmetry in amplitude. When the location of the underlying nerve is not certain, it is important to try several recording electrode positions to ensure that the maximal amplitude is obtained.

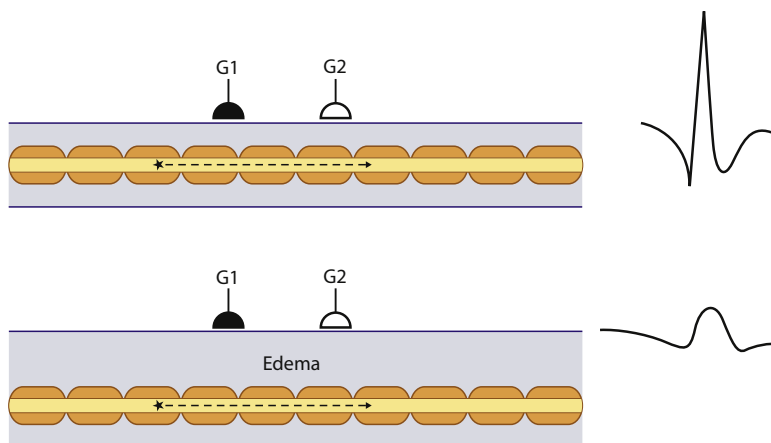


Fig. 8.28 Effect of increased distance between recording electrodes and nerve on amplitude. When performing sensory and mixed nerve conduction studies, the nerve is assumed to lie just under the skin (*top*). However, if edema is present, there will be a greater distance between the surface recording electrodes and the nerve (*bottom*). This results in a marked attenuation of the amplitude of the potential, and if the distance is great enough, the response can even be absent. In addition, the potential is dispersed in duration, the onset latency may be slightly shortened, and the peak latency may be slightly prolonged. This occurs because tissue acts as a high-frequency filter, attenuating the amplitude, which is predominantly a high-frequency response. The other changes occur from effects of volume conduction over a longer distance. Thus, caution must be exercised before interpreting any low or absent response as abnormal in the setting of marked edema, especially a sensory response.

potentials have a similar latency or, more importantly, if the sensory potential is absent, one might mistake the first component of the volume-conducted motor potential for the SNAP where none truly exists.

Distance Between Recording Electrodes and Nerve

In sensory or mixed nerve studies, the amount of intervening tissue and the distance separating the recording electrodes and the underlying nerve can markedly influence the amplitude of the recorded potential. As a potential is recorded at an increasing distance from the nerve, the amplitude decreases dramatically (Fig. 8.27). This accounts for the lower amplitude potentials seen with orthodromic sensory studies. In most orthodromic studies, the nerve lies deeper to the recording electrodes than it does in the corresponding antidromic study.

This situation is often encountered when performing lower extremity sensory studies (especially the sural and superficial peroneal sensory nerve studies) in a patient who has edema (Fig. 8.28). Regardless of the cause of edema (venous insufficiency and congestive heart failure being the most common), the edema results in a greater distance between the surface recording electrodes and the nerves than is normally seen. This then results in an attenuation of the amplitude. Thus, in this situation, caution must be exercised before interpreting any low or absent response, especially a sensory response, as abnormal. Indeed, in such a situation, it is only the presence of a normal response that is helpful. An absent or reduced response, in the presence of marked edema, should be noted in the report as possibly due to technical factors from the edema and should be appropriately incorporated into the final impression.

Lower amplitude potentials may be seen not only when the nerve lies deep but also when the recording

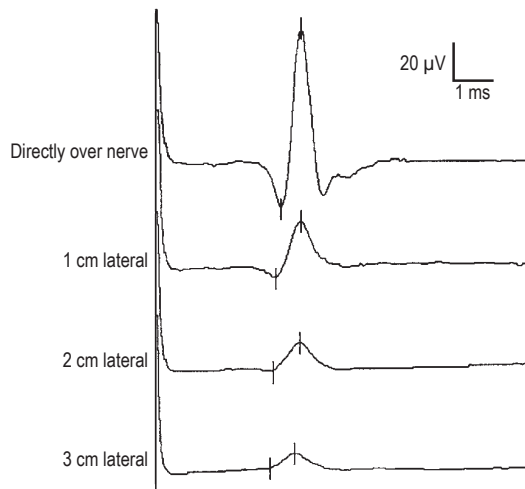


Fig. 8.29 Effect of distance between recording electrodes and nerve on latency. Median mixed nerve study, stimulating wrist, recording antecubital fossa. In addition to the effect on amplitude, if the recording electrodes are moved off the nerve while maintaining the same distance and stimulus current, the onset latency shifts to the left. This results in a spuriously fast conduction velocity. (Reprinted from Raynor EM, Preston DC, Logigian EL. Influence of surface recording electrode placement on nerve action potentials. *Muscle Nerve*. 1997;20:361, with permission of Wiley.)

electrodes are inadvertently placed lateral or medial to, and not directly over, the nerve. Because most nerves cannot be seen or palpated, recording electrodes for sensory and mixed nerve studies generally are placed based on anatomic landmarks and initially may not be placed in the optimal position directly over the nerve of interest. This situation occurs most frequently with sensory studies in which the position of the underlying nerve is slightly variable (e.g., palmar mixed studies, lateral antebrachial, medial antebrachial, sural, saphenous, and superficial peroneal sensory nerves). To avoid this pitfall, it is important to move the recording electrodes from the initial position slightly medially and then slightly laterally, with the stimulus current held constant, to determine which position yields the largest amplitude response. It often is surprising how minimal movement of the recording electrodes can greatly affect the amplitude of the response (Fig. 8.29). Failure to do so often can result in technical errors, especially when comparing amplitudes from side to side. The median and ulnar antidromic studies are an exception, as the recording electrodes are placed over the digits and one can always be assured that the recording electrodes are placed as close to the nerve as possible (i.e., directly over the digital nerves). The other exception is the superficial radial nerve, which can often be palpated as it runs over the extensor pollicis longus tendon. If one can palpate the nerve, the recording electrode can then be placed directly over it.

In addition to its effect on amplitude, the placement of the recording electrodes also affects the latency measurements. If the recording electrodes are placed lateral or

medial to the nerve, the onset latency shortens while the peak latency remains relatively unchanged. Although not intuitively obvious, these changes are due to the effects of volume conduction through tissue. The end result of the placement of recording electrodes at a distance from the nerve (because of intervening tissue, inaccurate placement of the electrodes, or both) is that the recorded electrical potential will be lower in amplitude and possibly spuriously fast (Fig. 8.29). The closer the recording electrodes are to the nerve, the higher the amplitude and the more accurate the onset latency.

Distance Between Active and Reference Recording Electrodes

Every potential recorded in a nerve conduction study is the result of the difference in electrical activity between the active and reference recording electrodes. For sensory and mixed nerve studies, the active and reference electrodes typically are placed in a straight line over the nerve to be recorded. Accordingly, the segment of nerve that is depolarized proceeds first under the active electrode and then passes distally to travel under the reference electrode. If the active and reference electrodes are too close together, they may briefly become electrically active at the same time, resulting in a lower amplitude potential due to a cancellation effect (Figs. 8.30 and 8.31). For this reason, the preferred inter-electrode distance between the active and reference recording electrodes for sensory and mixed nerve recordings is 3–4 cm. For the usual range of nerve conduction velocities, this distance ensures that depolarization will not occur under both electrodes simultaneously.

Limb Position and Distance Measurements

To compute a conduction velocity accurately, one must correctly measure the distance along the nerve. It usually is assumed that the surface distance accurately represents the true underlying length of the nerve, and in most circumstances that assumption is correct. There are several notable exceptions, however, the most important being that of the ulnar nerve across the elbow (Fig. 8.32). Surgical and cadaver dissection studies have shown that the ulnar nerve is slack and redundant when the arm is in the extended (i.e., straight) position. If surface distance measurements of the ulnar nerve are made with the arm extended, the true length of the underlying nerve is underestimated. Thus, ulnar nerve conduction studies performed with the elbow extended often result in artifactual slowing of conduction velocity across the elbow segment. When the elbow assumes a flexed position, the measured surface distance of the nerve across the elbow better reflects the true underlying length of the nerve, and a more valid measurement of nerve conduction velocity is made.

Surface distance measurements of several other nerves often are inaccurate. These include the radial nerve as it spirals around the humerus and the median and ulnar nerves

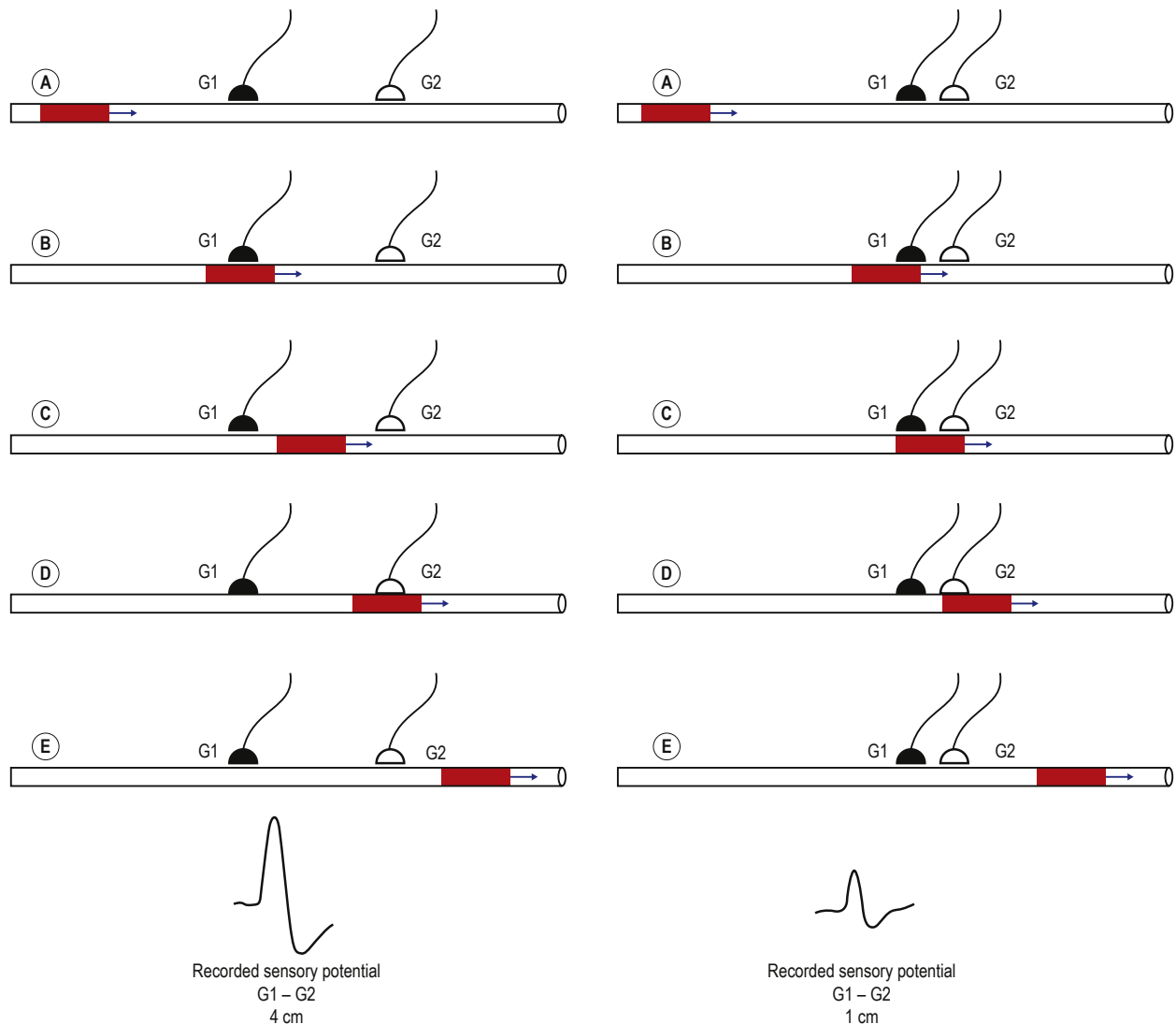


Fig. 8.30 Influence of distance between active and reference recording electrodes on sensory studies. The distance between the active (*G1*) and reference (*G2*) recording electrodes influences the morphology of the sensory nerve action potential (SNAP). The SNAP is the result of the difference in electrical activity between the active and reference recording electrodes. The segment of depolarized nerve proceeds first under the active electrode and then travels distally beneath the reference electrode (left side, inter-electrode distance of 4 cm). If the active and reference electrodes are too close (e.g., inter-electrode distance of 1 cm), they may briefly become electrically active at the same time, resulting in a lower-amplitude potential (right side, third trace). For the usual range of nerve conduction velocities in sensory and mixed nerve studies, separating the active and reference recording electrodes by 3–4 cm will ensure that depolarization does not occur under both electrodes simultaneously. Sites (A) to (E) show the progressive travel of the segment of nerve that is depolarized as it moves toward, under, and away from the recording electrodes.

between the axilla and Erb's point. In these situations, obstetric calipers can be used to more accurately approximate the true length of the underlying nerve.

Limb Position and Waveform Morphology

During any nerve conduction study where more than one site is stimulated (typically motor studies), it is essential that the limb remains in the same position for all stimulation sites. If this is not done, slightly different responses may result with different limb postures. This may occur due to slight movement of the skin (and recording electrodes)

in relation to the underlying muscle or nerve. In addition, there is the complicated issue of the “tendon potential” as discussed earlier. In the belly-tendon montage, it is generally assumed that the tendon is electrically inactive. However, this is not so for all nerves, especially the ulnar and tibial nerves, where the reference electrode placed over the tendon is often electrically active. Because there is no muscle over the tendon, this “tendon potential” is a volume-conducted far-field potential from proximal depolarizing muscles. These volume conducted potentials can change in shape and latency as the limb position changes. Thus take this example:

Fig. 8.31 Influence of distance between active and reference recording electrodes on sensory studies. Median sensory studies, stimulating the wrist, recording digit 2. The distance between the active ($G1$) and reference ($G2$) recording electrodes is 1.0 cm (*top*), 2.5 cm (*middle*), and 4.0 cm (*bottom*). Note the much smaller-amplitude potential when the recording electrodes are 1.0 cm apart. In this case, the active and reference electrodes are so close that the segment of depolarized nerve may occur simultaneously at both electrodes, resulting in a lower amplitude potential.

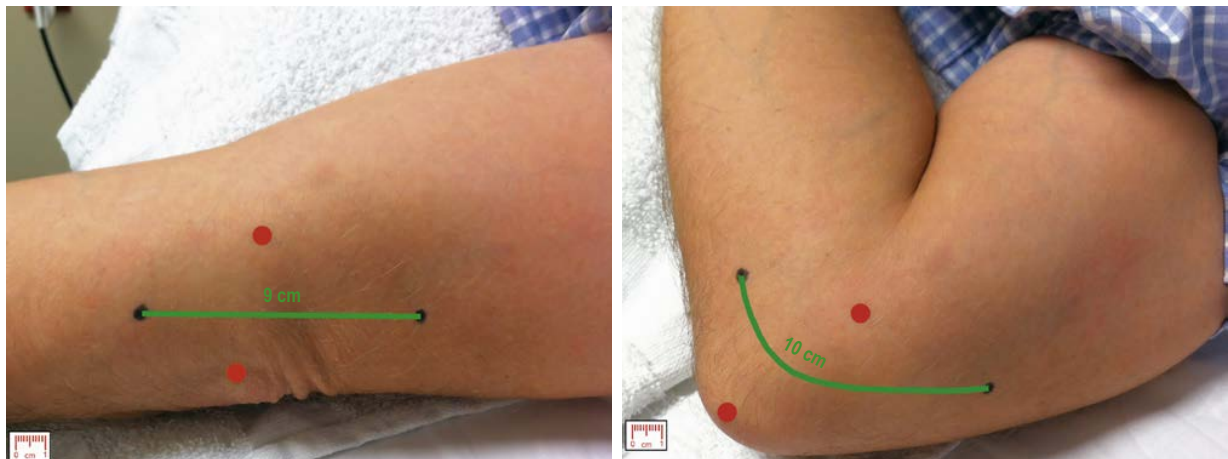
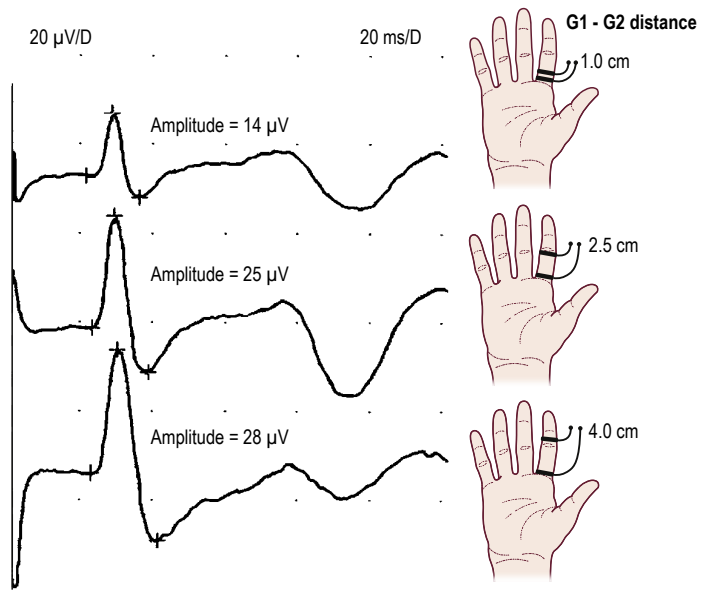


Fig. 8.32 Limb position and nerve length of the ulnar nerve. At the elbow, the ulnar nerve is slack and redundant when the arm is in the extended position. If surface distance measurements of the ulnar nerve are made with the arm in this position, the true length of the underlying nerve is underestimated. *Left*, With the elbow in extension, a surface distance of 9 cm is measured between the below- and above-elbow sites (note: the ulnar nerve runs between the medial epicondyle and olecranon marked by the red circles on the photos). *Right*, With the elbow in flexion, the same two marks now measure 10 cm apart, which more accurately reflects the true length of the ulnar nerve. If ulnar conduction studies are performed with the elbow extended, artifactual slowing of conduction velocity occurs across the elbow segment. When the elbow assumes a flexed position, the measured surface distance of the nerve across the elbow better reflects the true underlying length of the nerve, and a more valid measurement of nerve conduction velocity can be made.

- The ulnar nerve motor study is performed, stimulating the wrist, below-elbow, and above-elbow sites, with the arm in the bent (i.e., flexed) position for all three stimulation sites *vs.*
- The ulnar nerve motor study is performed, stimulating the wrist, below-elbow, and above-elbow sites. However, the ulnar nerve is stimulated at the wrist with the arm straight; then the elbow is flexed and the stimulations are done at the below-elbow and above-elbow sites

In this example, one would obtain slightly different amplitudes (especially at the below-elbow and above-elbow sites) and slightly different conduction velocities in the second scenario versus the first.

Although the physiology of volume conduction is complex and not intuitive, the bottom line is the following: if at all possible, during a nerve conduction study, stimulate all sites with the limb in the same position.

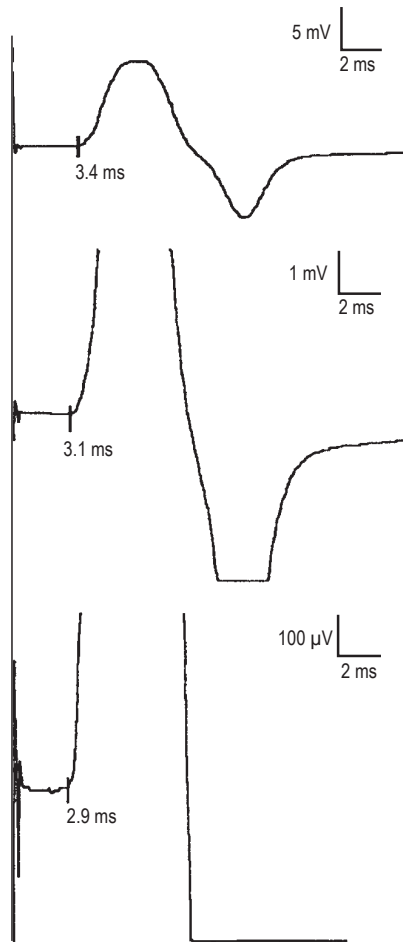


Fig. 8.33 Latency measurement and sensitivity. Median motor study, stimulating wrist, recording the abductor pollicis brevis, using varying sensitivities, with sweep speed held constant. Latency measurements should always be made using the same sensitivity. Note that as sensitivity is increased, latency measurement usually decreases.

Latency Measurements: Sweep Speed and Sensitivity

Both the sweep speed and sensitivity can markedly influence the recorded latency of both sensory and motor potentials. As the sensitivity is increased, the onset latency measurement tends to decrease (Fig. 8.33). Conversely, as the sweep speed is decreased, latency measurements usually increase (Fig. 8.34). For this reason, all latency measurements for each nerve conduction study should be made using the same sensitivity and the same sweep speed. This is especially true within nerves in which potentials obtained with different sweep speeds or sensitivities at distal and proximal stimulation sites along the nerve can easily result in the calculation of a faulty conduction velocity. This is one potential advantage of using peak latency as opposed to onset latency in sensory and mixed nerve studies, because peak latency is not affected by changes in either sweep speed or sensitivity (N.B., one cannot obtain a conduction velocity using peak latencies).

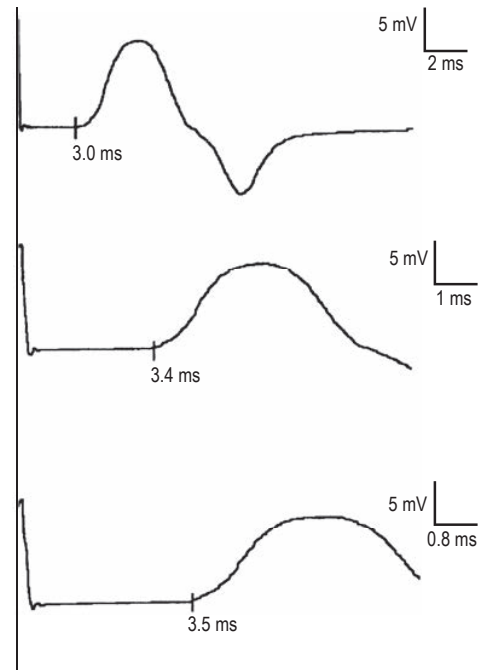


Fig. 8.34 Latency measurement and sweep speed. Median motor study, stimulating wrist, recording the abductor pollicis brevis, using varying sweep speeds, with sensitivity held constant. Latency measurements should always be made using the same sweep speed. Note that as sweep speed decreases, latency measurement usually increases.

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